

Operating Instructions

Diesel Engine with Bi-Fuel Kit

12V4000S83-E

12V4000S83LE

Application Group 4D

MS15651/00E



Power. Passion. Partnership.

Engine model	kW/cyl.	Application group
12V4000S83-E	139.8 kW/cyl.	4D, Frac operation (continuous operation at low load)
12V4000S83LE	155.4 kW/cyl.	4D, Frac operation (continuous operation at low load)

Table 1: Applicability

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1 Safety

1.1 Important provisions for all products

General

This product may pose a risk of injury or damage in the following cases:

- Incorrect use
- Operation, maintenance and repair by unqualified personnel
- Changes or modifications which are neither made nor authorized by the manufacturer
- Noncompliance with the safety instructions and warning notices

Nameplates

The product is identified by nameplate, model designation or serial number. This data must match the specifications in these instructions.

Nameplates, model designation or serial number can be found on the product.

All EU-certified engines delivered by MTU come with a second nameplate. When operating the machine in the EU: The second nameplate must be affixed in a prominent position as described in the accompanying specifications.

Emission regulations and emission labels

Responsibility for compliance with emission regulations

Modification or removal of any mechanical/electronic components or the installation of additional components including the execution of calibration processes that might affect the emission characteristics of the product are prohibited by emission regulations. Emission-related components must only be serviced, exchanged or repaired if the components used for this purpose are approved by the manufacturer.

Noncompliance with these specifications will invalidate the design type approval or certification issued by the emissions regulation authorities. The manufacturer does not accept any liability for violations of the emission regulations.

The product must be operated over its entire life cycle according to the conditions defined as "Intended use" (→ Page 7).

Emission certification applicable to engines with EPA Nonroad Tier 4 emission certification in accordance with 40 CFR 1039

Extract from the standard:

Failing to follow these instructions when installing a certified engine in a piece of nonroad equipment violates federal law (40 CFR 1068.105(b)), subject to fines or other penalties as described in the Clean Air Act.

Extract from the standard:

If you install the engine in a way that makes the engine's emission control information label hard to read during normal engine maintenance, you must place a duplicate label on the equipment, as described in 40 CFR 1068.105.

When fitting the second label, the requirements of 40 CFR 1068.105(c) must be followed and observed. This paragraph describes the process for requesting and fitting the label, the documentation obligations and storage obligations for the required documents.

Replacing components with emission labels

On all MTU engines fitted with emission labels, these labels must remain on the engine throughout its operational life.

Exception: Engines used exclusively in land-based, military applications other than by US government agencies.

Please note the following when replacing components with emission labels:

- The relevant emission labels must be affixed to the spare part.
- Emission labels shall not be transferred from the replaced part to the spare part.
- The emission labels must be removed from the replaced part and destroyed.

1.2 Correct use of all products

Correct use

The product is intended for use in accordance with its contractually-defined purpose as described in the relevant technical documents only.

Intended use entails operation:

- Within the permissible operating parameters in accordance with the (→ Technical data)
- With fluids and lubricants approved by the manufacturer in accordance with the (→ Fluids and Lubricants Specifications of the manufacturer)
- With preservation approved by the manufacturer in accordance with the (→ Preservation and Represervation Specifications of the manufacturer)
- With spare parts approved by the manufacturer in accordance with the (→ Spare Parts Catalog/MTU contact/Service partner)
- In the original as-delivered configuration or in a configuration approved by the manufacturer in writing (also applies to engine control/parameters)
- In compliance with all safety regulations and in adherence with all warning notices in this manual
- In compliance with the maintenance work and intervals specified in the (→ Maintenance Schedule) throughout the useful life of the product
- In compliance with the maintenance and repair instructions contained in this manual, in particular with regard to the specified tightening torques
- With the exclusive use of technical personnel trained in commissioning, operation, maintenance and repair

The product must not be operated in explosive atmospheres unless the engine fulfills the conditions for such use and approval has been granted.

Any other use, particularly misuse, is considered as being contrary to the intended purpose. Such improper use increases the risk of injury and damage when working with the product. The manufacturer shall not be held liable for any damage resulting from improper, non-intended use.

The specifications of the manufacturer will be amended or supplemented as necessary. Prior to operation, make sure that the latest version is used. The latest version can be found on the websites:

- For drive systems: <http://www.mtu-online.com>
- For power generation: <http://www.mtuonsiteenergy.com>

Modifications or conversions

Unauthorized changes to the product represent a contravention of its intended use and compromise safety.

Changes or modifications shall only be considered to comply with the intended use when expressly authorized by the manufacturer. The manufacturer shall not be held liable for any damage resulting from unauthorized changes or modifications.

1.3 Personnel and organizational requirements

Organizational measures of the user/manufacturer

This manual must be issued to all personnel involved in operation, maintenance, repair, assembly, installation, or transportation.

Keep this manual handy in the vicinity of the product such that it is accessible to operating, maintenance, repair, assembly, installation, and transport personnel at all times.

Personnel must receive instruction on product handling and repair based on this manual. In particular, personnel must have read and understood the safety requirements and warnings before starting work.

This is important in the case of personnel who only occasionally perform work on or around the product. Such personnel must be instructed repeatedly.

Personnel requirements

All work on the product must be carried out by trained, instructed and qualified personnel only:

- Training at the Training Center of the manufacturer
- Qualified personnel from the areas mechanical engineering, plant construction, and electrical engineering

The operator must define the responsibilities of the personnel involved in operation, maintenance, repair, assembly, installation, and transport in writing.

Personnel shall not report for duty under the influence of alcohol, drugs or strong medication.

Clothing and personal protective equipment

Always wear appropriate personal protective equipment, e.g. safety shoes, ear protectors, protective gloves, goggles, breathing mask. Follow the instructions concerning personal protective equipment in the descriptions of the individual activities.

Safe handling of Substances of Very High Concern pursuant to the REACH regulation (Registration, Evaluation, Authorisation and restriction of CHemicals): We recommend wearing protective gloves at all times in order to reduce risk when working.

1.4 Transport

Transport

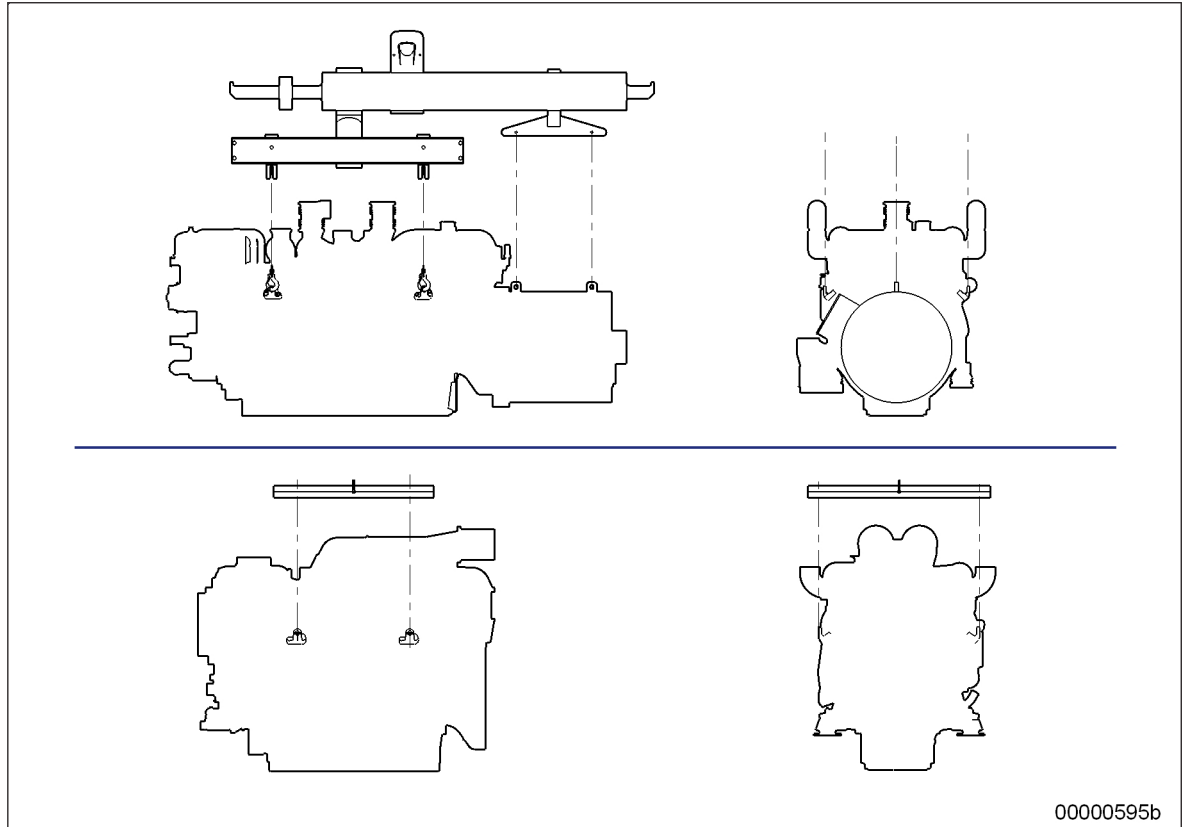


Figure 1: Transport

Always use the lifting eyes on the engine and generator/gearbox when transporting gensets.

Always use the lifting eyes on the engine when transporting an engine separately.

Only use transport and lifting devices approved by MTU.

The engine/genset must only be transported in installation position: maximum permissible diagonal pull 10°.

Remove any loose parts on the genset.

Raise the engine/genset slowly. The lifting ropes or chains must not make contact with the engine or its components. Readjust lifting device as necessary.

Pay attention to the center of gravity of the engine/genset: Refer to installation drawing.

For special packaging with aluminium foil: Suspend the engine/genset by the lifting eyes on the bearing pedestal or transport by means of handling equipment (forklift truck) capable of bearing the load.

Fit the crankshaft transportation lock on the engine and fit the engine mount locking devices prior to transport.

Secure the engine/genset such as to preclude tipping during transport. Secure such as to preclude slipping and tipping when driving up or down inclines and ramps.

Placement after transport

Place the engine/genset on a firm, flat surface only.

Make sure that the consistency and load-bearing capacity of the ground or support surface is adequate.

Never set an engine down on the oil pan unless expressly authorized to do so by MTU on a case-to-case basis.

1.5 Crankshaft transport locking device

Special tools, Material, Spare parts

Designation / Use	Part No.	Qty.
Torque wrench, 10–60 Nm	F30452769	1
Torque wrench, 60–320 Nm	F30452768	1
Engine oil		

Note: The locking device protects the crankshaft bearings from shocks and vibration damage during engine transport.

For installation and removal of the transport locking device, follow the instructions below:

1. The transport locking device must remain installed as long as possible during engine installation in order to avoid damage.
2. Prior to every engine transport, the transport locking device must be reinstalled on both sides according to the instructions.
3. If the engine is to be moved together with the generator, the transport locking device for the generator must also be installed.
4. Always use the screws supplied with or installed in the transport locking device to secure it on the engine.
5. Starting or barring the engine is allowed only with the transport locking device removed. If the generator/transmission is already mounted on the engine, ensure that the transport locking device of the generator/transmission is also removed.

Removing the transport locking device from driving end (KS), variant A

1. Remove parts (1) to (8) from engine. Spacer sleeves (5) are optional.
2. Store the removed parts of the transport locking device carefully for possible reuse.

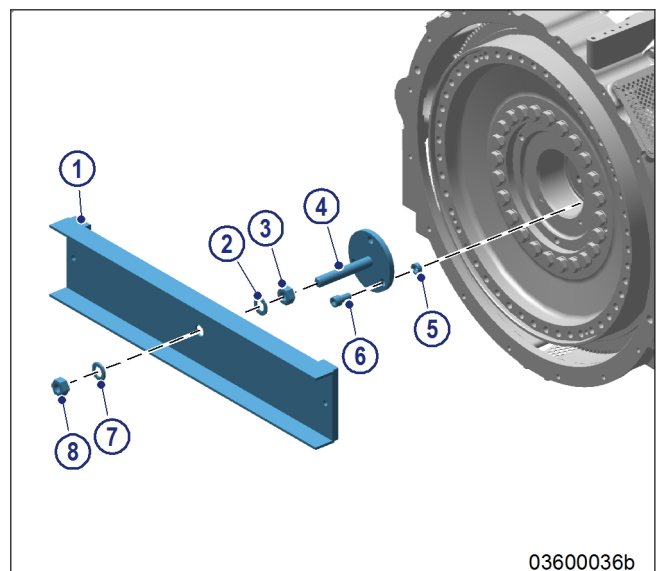


Figure 2: Removing the transport locking device, variant A